

Lecture(19)

Chapter(6)

Graph Theory

- Introduction
- Graphs and Multigraphs
- Finite Graphs & Trivial Graph
- Subgraphs & Isomorphic Graph

Introduction

○ **What is a graph?**

We begin by considering Figs. 1.1 and 1.2, which depict part of a road map and part of an electrical network.

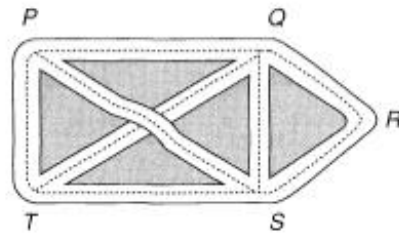


Fig. 1.1

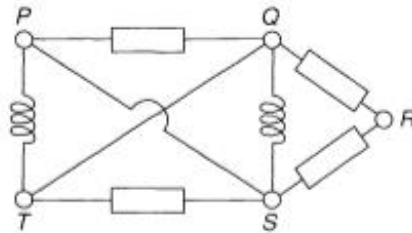


Fig. 1.2

Introduction

- Either of these situations can be represented diagrammatically by means of points and lines, as in Fig. 1.3. The points **P**, **g**, **R**, **S** and **T** are called vertices, the **lines** are called edges, and the whole diagram is called a graph.

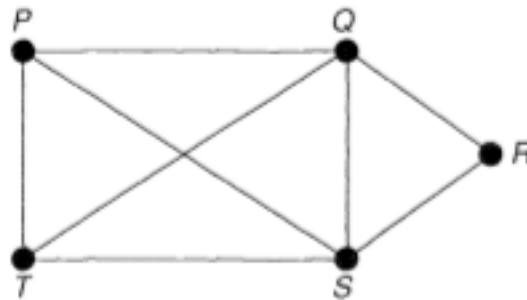
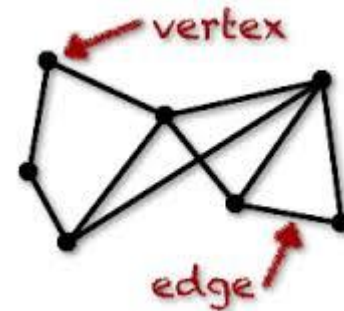


Fig. 1.3

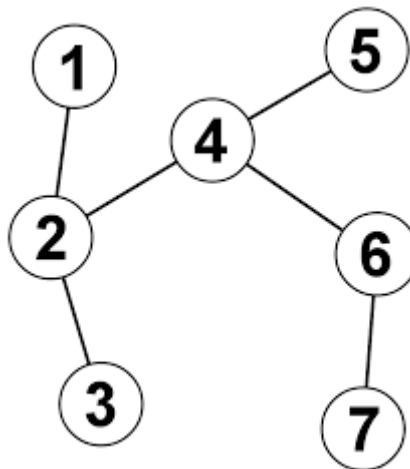


- Note that the intersection of the lines **PS** and **QT** is not a vertex, since it does not correspond to a cross-roads or to the meeting of two wires.
- Thus, a graph is a representation of a set of points and of how they are joined up.

Introduction

- Informally, a graph is a diagram consisting of points, called vertices, joined together by lines, called edges; **each edge joins exactly two vertices.**
- A graph G is a triple consisting of a vertex set of $V(G)$, an edge set $E(G)$.

Example:

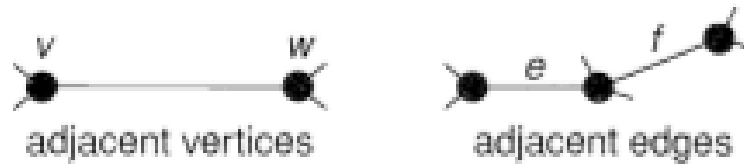


- $V := \{1, 2, 3, 4, 5, 6, 7\}$
- $E := \{\{1,2\}, \{2,4\}, \{2,3\}, \{4,5\}, \{4,6\}, \{6,7\}\}$

Introduction

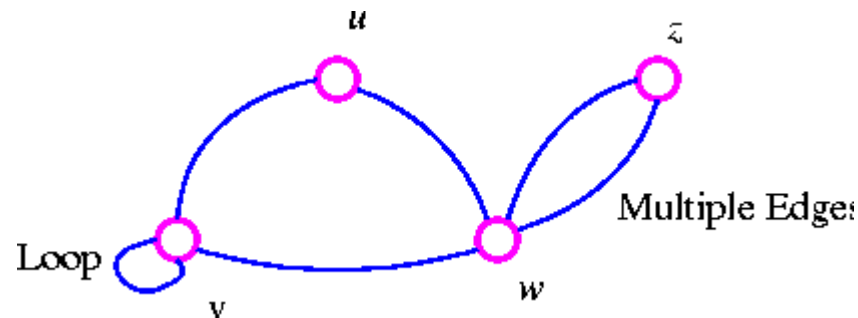
○ Adjacency

We say that two vertices v and w of a graph G are **adjacent** if there is an edge vw joining them, and the vertices v and w are then **incident** with such an edge. Similarly, two distinct edges e and f are adjacent if they have a vertex in common. The vertices v and w are called **endpoints** of the edge $\{v, w\}$.



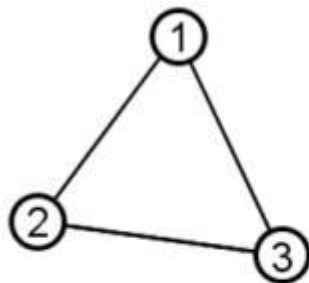
○ Loop and Multiple Edges

A loop  is an edge whose endpoints are equal i.e., an edge joining a vertex to it self is called a loop. We say that the graph has multiple edges if in the graph two or more edges joining the same pair of vertices.



Undirected and Directed Graphs

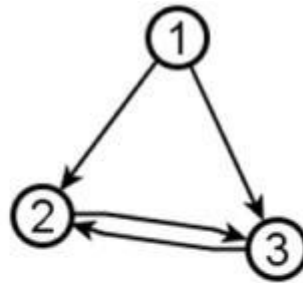
- **Undirected graph:** The edges of a graph are assumed to be **unordered pairs of nodes**. Sometimes we say *undirected* graph to emphasize this point. In an undirected graph, **we write edges using curly braces to denote unordered pairs**. For example, an undirected edge $\{2,3\}$ from vertex 2 to vertex 3 is the same thing as an undirected edge $\{3,2\}$ from vertex 3 to vertex 2.
- **Directed graph:** In a *directed* graph, the two directions are counted as being distinct *directed edges*. In an directed graph, **we write edges using parentheses to denote ordered pairs**. For example, edge $(2,3)$ is directed from 2 to 3, which is different than the directed edge $(3,2)$ from 3 to 2. **Directed graphs are drawn with arrowheads** on the links, as shown below:



Undirected graph (V_1, E_1)

$$V_1 = \{1, 2, 3\}$$

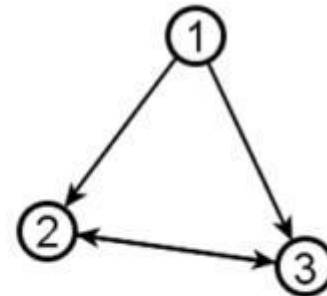
$$E_1 = \{\{1, 2\}, \{2, 3\}, \{3, 1\}\}$$



Directed graph (V_2, E_2)

$$V_2 = \{1, 2, 3\}$$

$$E_2 = \{(1, 2), (2, 3), (3, 2), (1, 3)\}$$



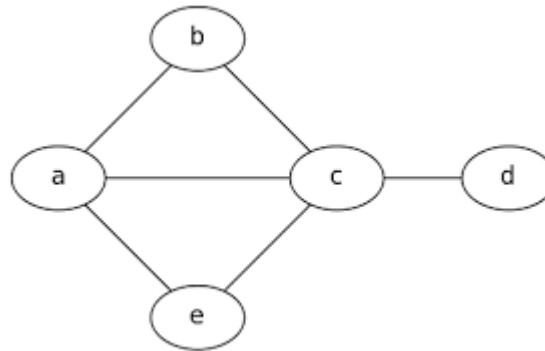
Easier way to draw

directed graph (V_2, E_2)

Simple Graphs and Multigraphs

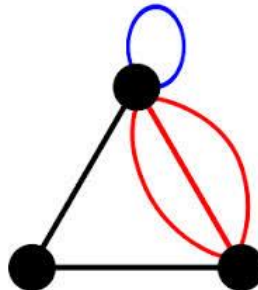
○ Simple Graphs

Simple graphs are graphs without multiple edges or self-loops.



○ Multigraph

A *multigraph*, as opposed to a simple graph, is an undirected graph in which multiple edges (and sometimes loops) are allowed.

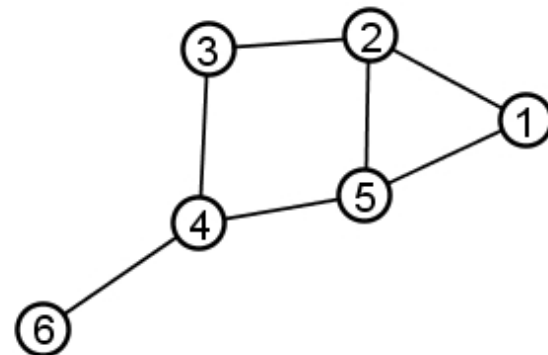


Neighborhood and Degree

- Two vertices are called **adjacent** if they share a common edge, in which case the common edge is said to **join** the two vertices. An edge and a vertex on that edge are called **incident**.
- See the 6-node graph Fig 1.4 for examples of **adjacent** and **incident**:
 - Nodes 4 and 6 are adjacent (as well as many other pairs of nodes)
 - Nodes 1 and 3 are not adjacent (as well as many other pairs of nodes)
 - Edge $\{2,5\}$ is incident to node 2 and node 5.
- The **neighborhood** of a vertex v in a graph G is the set of vertices adjacent to v . The neighborhood is denoted $N(v)$. The neighborhood does not include v itself. For example, in the graph below $N(5) = \{4,2,1\}$ and $N(6) = \{4\}$.
- The **degree** of a vertex is the total number of vertices adjacent to the vertex. The degree of a vertex v is denoted $\deg(v)$. We can equivalently define the degree of a vertex as the cardinality of its neighborhood and say that for any vertex v , $\deg(v) = |N(v)|$.

Vertex	Degree
1	2
2	3
3	2
4	3
5	3
6	1

Fig 1.4



Degree of a Vertex

Theorem: The sum of the degrees of the vertices of a graph G is equal to twice

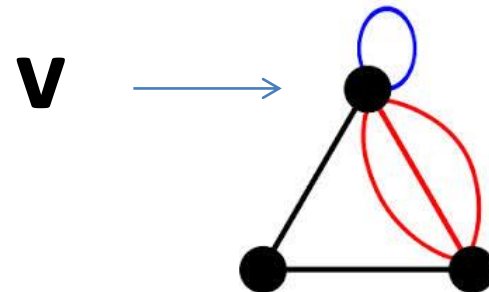
the number of edges in G , i.e. $2e = \sum_{v \in V} \deg(v)$

Illustration: Consider the graph in Fig 1.4. The sum of degrees equals 14 which, as expected, is twice the number of edges.

Example: How many edges are there in a graph with 10 vertices each of degree 6?

Solution: It follows that $2e = 60$. Therefore, $e = 30$.

- A vertex is said to be **even** or **odd** according as its degree is an even or an odd number. Thus 1 and 3 are even whereas 2, 4, 5 and 6 are odd vertices in Fig 1.4.
- The Theorem also holds for multigraphs where a loop is counted twice toward the degree of its endpoint. For example, in the below graph we have $\deg(V) = ?$ Why.



- A vertex of degree zero is called an **isolated** vertex.

Finite Graphs & Trivial Graph

- A **finite graph** is a graph in which the vertex set and the edge set are finite sets. Otherwise, it is called an **infinite graph**.
- Most commonly in graph theory it is implied that the graphs discussed are finite. If the graphs are infinite, that is usually specifically stated.
- The finite graph with one vertex and no edges, i.e., a single point, is called the **trivial graph**.

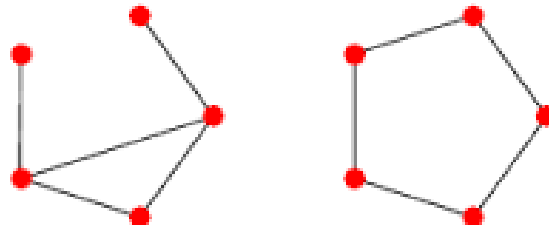
$n = 1$



$n = 4$



$n = 5$



Subgraphs Graphs

- A **subgraph** of a graph G is a graph, each of whose vertices belongs to $V(G)$ and each of whose edges belongs to $E(G)$. Thus the graph in Fig. 2.13 is a subgraph of the graph in Fig. 2.14, but is not a subgraph of the graph in Fig. 2.15, since the latter graph contains no 'triangle'.

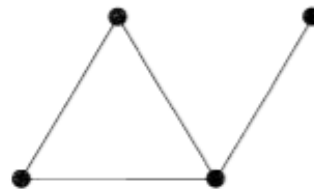


Fig. 2.13

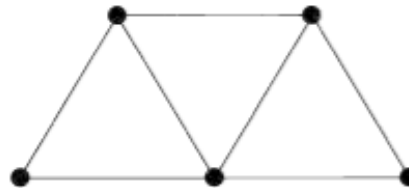


Fig. 2.14

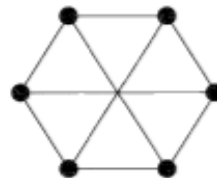


Fig. 2.15

Subgraphs Graphs

- We can obtain subgraphs of a graph by deleting edges and vertices. If e is an edge of a graph G , we denote by $G - e$ the graph obtained from G by deleting the edge e . More generally, if F is any set of edges in G , we denote by $G - F$ the graph obtained by deleting the edges in F . Similarly, if v is a vertex of G , we denote by $G - v$ the graph obtained from G by deleting the vertex v together with the edges incident with v . More generally, if S is any set of vertices in G , we denote by $G - S$ the graph obtained by deleting the vertices in S and all edges incident with any of them. Some examples are shown in Fig. 2.16.

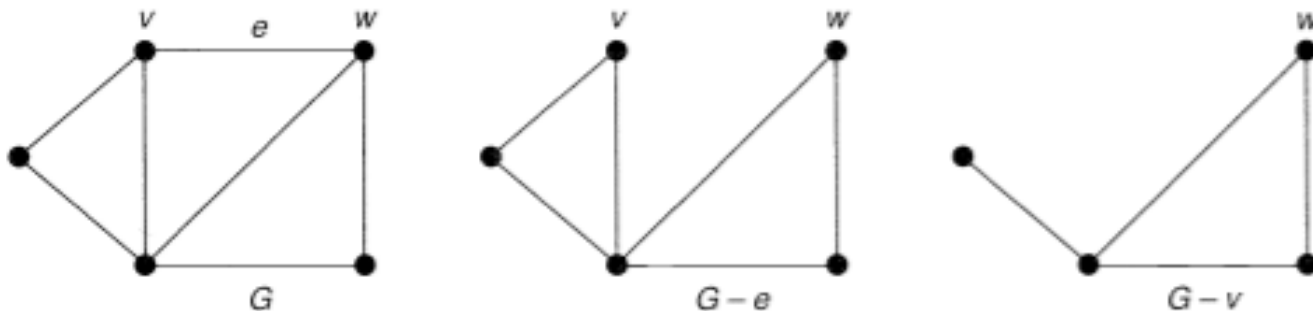


Fig. 2.16

Isomorphic Graphs

- Two simple graphs G and H are isomorphic if there is a bijection $Q: V(G) \rightarrow V(H)$ which preserves adjacency and nonadjacency $uv \in E(G) \Leftrightarrow Q(u)Q(v) \in E(H)$
- In another words, Two graphs G_1 and G_2 are **isomorphic** if there is a one-one correspondence between the vertices of G_1 and those of G_2 such that the number of edges joining any two vertices of G_1 is equal to the number of edges joining the corresponding vertices of G_2 . Thus the two graphs shown in Fig. 2.3 are isomorphic under the correspondence $u \leftrightarrow l, v \leftrightarrow m, w \leftrightarrow n, x \leftrightarrow p, y \leftrightarrow q, z \leftrightarrow r$

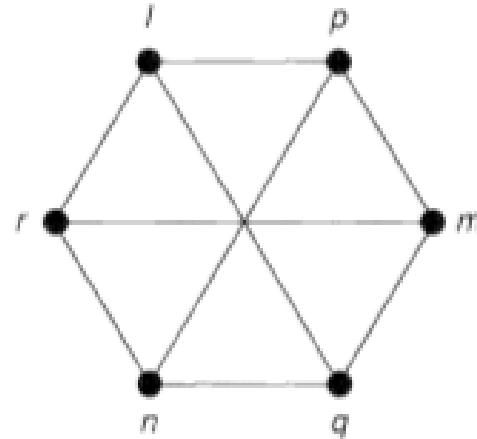
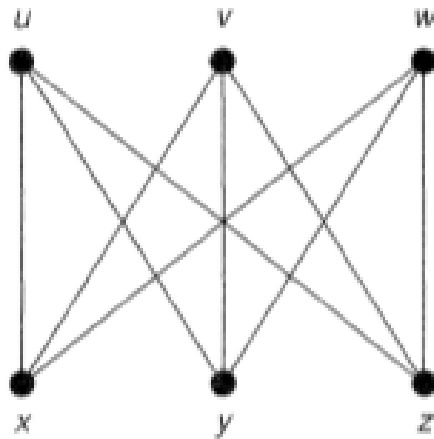


Fig. 2.3

Isomorphic Graphs

Example: (a) Show that the graphs $G(U, E)$ and $H(V, F)$ are isomorphic in Fig 2.17.
(b) show that the graphs displayed in Fig 2.18 are not isomorphic.

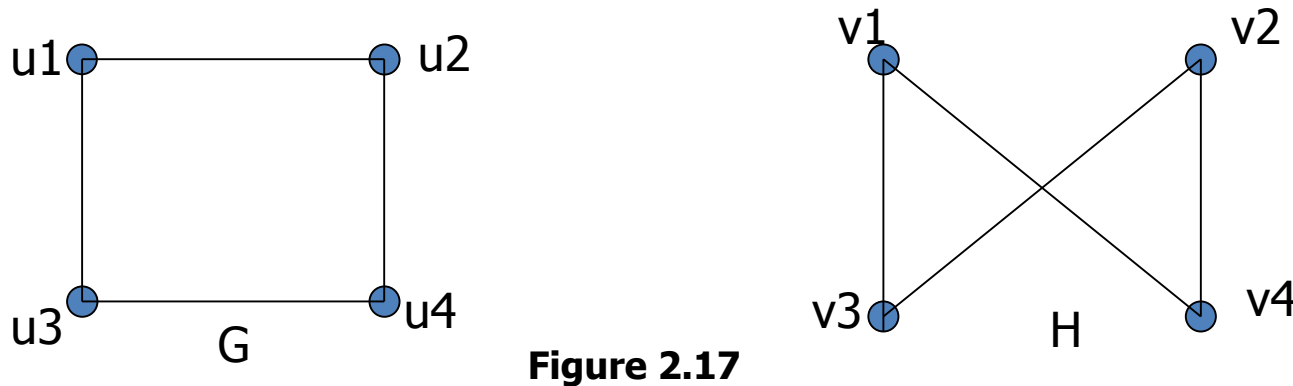


Figure 2.17

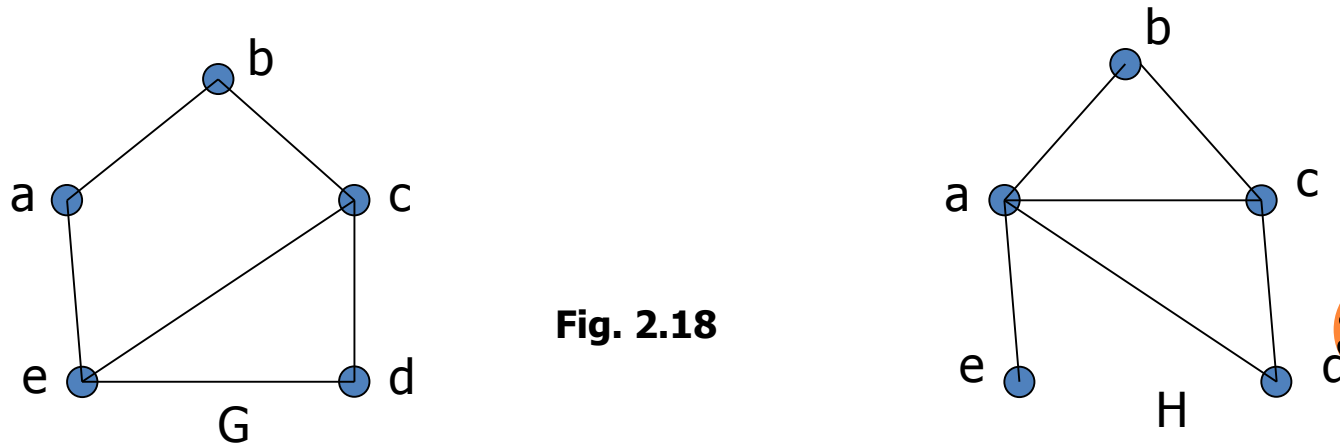


Fig. 2.18

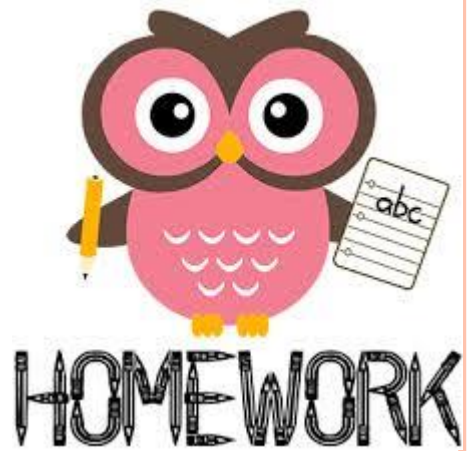
Isomorphic Graphs

Solution:

(a) The function f with $f(u_1) = v_1$, $f(u_2) = v_4$, $f(u_3) = v_3$, and $f(u_4) = v_2$ is a one-to-one correspondence between V and W . We see that this correspondence preserves adjacency.

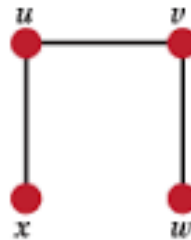
Solution:

(b) Both G and H have five vertices and six edges. However, H has a vertex of degree 1, namely e , whereas G has no vertices of degree 1. It follows that G and H are not isomorphic.



Determine whether or not the following pairs of graphs are isomorphic.

a)



b)

